

Computational supplement for Residual cohomology and causal transport in free pure-Weyl gravity

GPT-5.6.sol*

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Abstract

This versioned supplement records the computational claim boundaries, generated-input inventory, independent checks and clean-release procedure for the two standalone articles `conformal-residual-cohomology-krein.tex` and `conformal-covariant-causal-transport.tex`. The former contains the residual cohomology theorem and the latter the Lorentzian causal transport theorem. The combined source `conformal-residual-cohomology.tex` is retained as an archival cross-reference, not as the primary submission. Machine receipts support the articles but do not replace their mathematical arguments or independent expert review.

1 Scope and publication split

The section-level extraction map and submission gates are maintained in `../notes/conformal-paper-split-roadmap.md`. This supplement serves two articles:

1. the residual cohomology and pairing calculation;
2. the support-local causal curvature/tractor bridge which transports that result to covariant cohomology.

It contains implementation receipts, not a third competing theorem statement. Paper A's headline claim is tagged `LOCAL-ALGEBRAIC`; Paper B's causal bridge is tagged `LORENTZIAN-CAUSAL`. Reduced-mode and Euclidean checks are regressions only where the causal article says so.

2 Focused article dependency ledgers

Paper A: residual cohomology and pairing

Claim	Authoritative executable or receipt
Weyl module and exact generator action	<code>symbolic/verify_conformal_generator_all_levels.py</code> , <code>symbolic/verify_conformal_energy_mode_krein.py</code> .

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

Claim	Authoritative executable or receipt
Relative weight-four kernel	<code>symbolic/verify_conformal_relative_brst_weight4.py</code> and the independent rank-53 check below.
Absolute one-particle acyclicity and two-particle rank	<code>symbolic/verify_conformal_global_brst_window.py</code> .
Cartan reduction and completed residual theorem	<code>symbolic/verify_conformal_cartan_contraction.py</code> , <code>analytic_completion/certificates/completed_H4.json</code> .
Residual Gram matrix and selected BV–BFV transfer	<code>analytic_completion/certificates/completed_gram.json</code> and the <code>field_bv_identification/</code> certificate chain.

Paper A computes vertex/deformation classes in the selected residual polarization. It does not claim a positive graviton Hilbert space or a Lorentzian off-shell propagator.

Paper B: covariant causal transport

Claim	Authoritative executable or receipt
Curved auxiliary identities, local retract and current comparison	<code>symbolic/verify_conformal_covariant_completion.py</code> and the curved certificate chain indexed by the proof ledger.
Weyl–Cotton closure and sourced constraint propagation	<code>symbolic/verify_conformal_weyl_cotton_causal_pde.py</code> and <code>covariant_completion/certificates/curved_weyl_cotton_hyperbolic.json</code> .
All-row local prolongation and $356 + 30$ cyclic retract	<code>symbolic/verify_conformal_full_prolonged_green_homotopy_assembly.py</code> .
Adjoint-tractor causal homotopies	<code>symbolic/verify_conformal_adjoint_tractor_green_transfer.py</code> and <code>covariant_completion/certificates/adjoint_tractor_green_transfer.json</code> .
Causal quasi-isomorphism, endpoint recovery and equivariance	<code>symbolic/verify_conformal_final_covariant_transport.py</code> .
Pairing transport and final H_{cov}^4 theorem	<code>covariant_completion/generated/covariant_H4_proof_ledger.md</code> .

Paper B transports Paper A’s residual calculation; it does not perform a second finite-cutoff CE computation. The false legacy flags for a canonical endpoint inverse and monolithic prolonged witness are not dependencies of the direct tractor causal route.

The status file `covariant_completion/certificates/curved_curvature_prolongation_status.json` also preserves historical false subflags from the superseded rank-14 equation-cone, direct-Weyl–Cotton inverse, and first-order search routes. They are intentionally not promoted by the successful tractor construction. For the publication theorem the authoritative dependency set is the 23-requirement terminal DAG in `covariant_completion/generated/covariant_H4_proof_ledger.md`; its true `direct_tractor_causal_homotopy` and `causal_green_homotopy` gates do not imply that the legacy objects now exist.

3 Independent central residual-rank check

The command

```
python3 symbolic/verify_conformal_residual_rank53_independent.py
```

uses only the Python standard library. It constructs the integral spin-two $\mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ action on

$$\mathrm{Sym}^2((2, 0) \oplus (0, 2)), \quad 55 = 15 + 25 + 15,$$

and obtains exact block ranks

$$53 = 14 + 25 + 14.$$

The two exact kernel vectors are the chiral binary-quartic singlets

$$e_0 e_4 - 4e_1 e_3 + 3e_2^2.$$

Ranks are computed over two prime fields. Since modular rank is a characteristic-zero lower bound and the two displayed independent exact kernel vectors give the complementary upper bound, the characteristic-zero rank is exactly 53.

This independent check is intentionally narrow. It does not replace the one-particle $290 \rightarrow 1311 \rightarrow 3657$ absolute window, full Chevalley–Eilenberg nilpotency, BV–BFV transfer or covariant causal certificates.

4 Generated-input inventory

The archival paper has one active generated TeX input. It is tracked by Git and its header binds the source certificate digest:

```
generated/endpoint_factorization_nullstellensatz.tex.
```

The clean-release audit recursively discovers every `\input` and `\include` in Paper A, Paper B, this supplement and the archival monolith. It rejects an input absent from the selected Git tree.

Other generated TeX files in `analytic_completion/generated/` and `covariant_completion/generated/` are tracked standalone receipts, not active `\input` dependencies. Their classification is:

- *current theorem receipts*: `completed_residual.tex`, `energy_mode_krein.tex`, `cauchy_sobolev.tex`, `curl_factorization.tex`, and `minimal_ghost_witness.tex`;
- *scoped historical receipts*: `auxiliary_green_witness.tex` and `covariant_bv_last_mile_status.tex`, which record superseded canonical-witness checkpoints and are not current terminal gates;
- *active generated input*: the Nullstellensatz receipt included below, whose generator is checked for byte-for-byte freshness.

5 Endpoint factorization receipt

For continuity with the archival appendix, the active generated input is included below. It is a scoped no-go receipt for the complete parallel invariant two-wave-factor family; it is not used as evidence against the direct tractor causal homotopy.

$$\begin{aligned}
 f_0 &= -2a_0a_4 - a_0a_5 \\
 &\quad - 16a_1a_4 - 8a_1a_5 \\
 &\quad - 8a_2a_4 - 4a_2a_5 \\
 &\quad + 2 \\
 w_0 &= \frac{3}{128} \left(18a_0^2 - 192a_0a_1 \right. \\
 &\quad - 72a_0a_2 + 23a_0a_6 \\
 &\quad + 384a_1^2 + 192a_1a_2 \\
 &\quad + 62a_1a_6 + 58a_2a_6 \\
 &\quad + a_3a_7 - 152a_4a_6 \\
 &\quad + 224a_4a_8 - 68a_5a_6 \\
 &\quad \left. + 112a_5a_8 \right)
 \end{aligned}$$

$$\begin{aligned}
 f_1 &= 2a_0a_6 - 3a_0a_8 \\
 &\quad + 16a_1a_6 - 24a_1a_8 \\
 &\quad + 8a_2a_6 - 12a_2a_8 \\
 &\quad - 2 \\
 w_1 &= \frac{3}{128} \left(27a_0a_4 + 24a_0a_5 \right. \\
 &\quad - 48a_1a_4 - 60a_1a_5 \\
 &\quad + 2a_3^2 + 2a_3a_7 \\
 &\quad - 224a_4^2 - 224a_4a_5 \\
 &\quad - 56a_5^2 + 6a_7^2 \\
 &\quad \left. - 58 \right)
 \end{aligned}$$

$$\begin{aligned}
 f_2 &= -a_0a_8 + 6a_1a_6 \\
 &\quad - 8a_1a_8 + 2a_2a_6 \\
 &\quad - 4a_2a_8 - 2 \\
 w_2 &= -\frac{3}{128} \left(81a_0a_4 + 72a_0a_5 \right. \\
 &\quad - 144a_1a_4 - 180a_1a_5 \\
 &\quad + 6a_3^2 + 6a_3a_7 \\
 &\quad - 224a_4^2 - 224a_4a_5 \\
 &\quad - 56a_5^2 + 18a_7^2 \\
 &\quad \left. - 174 \right)
 \end{aligned}$$

$$\begin{aligned}
 f_4 &= -2a_0a_3 \\
 w_4 &= \frac{3}{128} \left(2a_3a_6 - a_4a_7 \right. \\
 &\quad \left. + a_5a_7 - 4a_6a_7 \right)
 \end{aligned}$$

$$\begin{aligned}
 f_7 &= -\frac{1}{3} \left(3a_0a_5 + 18a_1a_4 \right. \\
 &\quad + 6a_1a_5 + 6a_2a_4 \\
 &\quad \left. + 6a_2a_5 - 10 \right) \\
 w_7 &= \frac{3}{128} \left(21a_0a_6 - 72a_1a_6 \right. \\
 &\quad - 3a_3a_7 - 64a_4a_6 \\
 &\quad \left. - 32a_5a_6 \right)
 \end{aligned}$$

$$\begin{aligned}
f_9 &= 2a_7(a_1 - a_2) \\
w_9 &= -\frac{3}{128}(a_3a_4 - a_3a_5 \\
&\quad - 6a_6a_7) \\
f_{14} &= -48a_1^2 - 48a_1a_2 \\
&\quad - 12a_2^2 + 4a_4a_6 \\
&\quad - 6a_4a_8 + 2a_5a_6 \\
&\quad - 3a_5a_8 - 1 \\
w_{14} &= -\frac{1}{64}(3a_0^2 - 288a_0a_1 \\
&\quad - 108a_0a_2 - 135a_0a_4 \\
&\quad - 36a_0a_5 - 174a_1^2 \\
&\quad - 516a_1a_2 + 432a_1a_4 \\
&\quad + 108a_1a_5 - 174a_2^2 \\
&\quad - 110) \\
f_{15} &= -16a_1^2 - 16a_1a_2 \\
&\quad - 4a_2^2 - 2a_4a_8 \\
&\quad - a_5a_8 - 1 \\
w_{15} &= \frac{9}{64}(a_0^2 - 96a_0a_1 \\
&\quad - 36a_0a_2 - 21a_0a_4 \\
&\quad - 58a_1^2 - 172a_1a_2 \\
&\quad + 80a_1a_4 + 4a_1a_5 \\
&\quad - 58a_2^2 - 26) \\
f_{17} &= -a_0^2 + 6a_4a_6 \\
&\quad - 8a_4a_8 + 2a_5a_6 \\
&\quad - 4a_5a_8 + 4 \\
w_{17} &= -\frac{3}{64}(18a_0a_4 + 9a_0a_5 \\
&\quad - 48a_1a_4 - 24a_1a_5 \\
&\quad - 20) \\
f_{18} &= -\frac{1}{2}(2a_0^2 - 2a_3^2 \\
&\quad - a_4a_6 + a_5a_6 \\
&\quad - 4) \\
w_{18} &= -\frac{3}{16} \\
f_{19} &= -a_6(a_3 + a_7) \\
w_{19} &= \frac{3}{32}(3a_0a_7 - a_1a_3 \\
&\quad + a_2a_3) \\
f_{25} &= -\frac{1}{2}(2a_1^2 - 4a_1a_2 \\
&\quad + 2a_2^2 - a_4a_6 \\
&\quad + a_5a_6 - 2a_7^2 \\
&\quad + 4) \\
w_{25} &= -\frac{9}{16}
\end{aligned}$$

With all omitted multipliers equal to zero, direct expansion gives

$$\sum_{i \in \{0,1,2,4,7,9,14,15,17,18,19,25\}} w_i f_i = 1.$$

The multiplier degrees are at most two. Thus the residual order-two ideal is the unit ideal over $\mathbb{Q}[a_0, \dots, a_8]$.

6 Repository computational material

Block	Versioned computational content
Finite-jet convention audit	Coordinate bases, globalization coverage and independent convention checks.
Residual ranks	Sparse formats, good-prime embeddings, pivot/minor witnesses, exact kernel representatives and the independent rank-53 check.
BV–BFV dictionaries	Raw/minimal/gauge-fixed basis ledgers, zero-mode sections, contraction maps and sign conventions.
Curvature prolongation	Weyl–Cotton row tables, sourced subsidiary system, mapping-cylinder rows, PBW normal forms and cyclic projectors.
Alternative witnesses	Scoped scalar-wave, first-order, factorization and relative-saddle searches, including exact negative receipts.
Publication provenance	Certificate schemas, content hashes, terminal proof ledger, commands, timeouts and artifact policy.

7 Clean release and reproduction

From the repository project directory, the isolated release audit is

```
python3 symbolic/audit_conformal_publication_release.py \
  --tree-ish HEAD \
  --receipt conformal-publication-release-audit.json
```

The selected tree-ish must already contain the four TeX sources and four PDFs. Untracked or merely staged working-tree files are deliberately ignored, so this command is a post-commit release gate rather than a development-tree preview. It uses `git archive` rather than the development working tree, verifies that all four TeX entrypoints, all four publication PDFs and every recursively active TeX input are tracked in the selected tree, checks the generated Nullstellensatz input, builds each document to reference stability, and runs the independent rank, provenance and proof-ledger checks. The JSON receipt is a CI artifact, not an authoritative mathematical input.

The focused clean-release audit complements, rather than replaces, the full theorem rail

```
python3 symbolic/verify_conformal_paper_free.py --reproduce --timeout 1800
```

and the content manifest check

```
python3 symbolic/update_conformal_paper_snapshot.py --check
```

The release audit is fail-closed on missing tracked entrypoints, unresolved TeX inputs, stale generated TeX, failed independent checks, failed stabilized builds, undefined references and stale manifests. The archival equation labels remain available for referee auditability even though Papers A and B are now the primary reading entrypoints.